CLI Use

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1. Course Content

- Learn about common Ultralytics CLI command tools.

2. Case Demonstration

CLI is a way for users to interact with a computer or software program, in which programs are run or tasks are performed by typing text commands. Ultralytics provides a very convenient command-line interaction tool.

2.1 Image Prediction

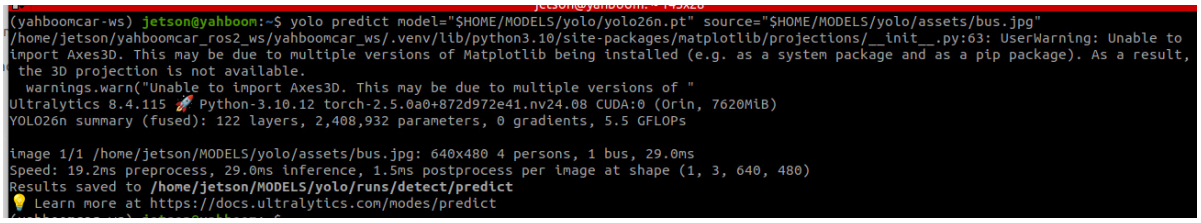
- First enter the yolo environment

> **Note**
>
> - To prevent mutual pollution and interference between Python environments, the yolo library files are located in the directory `~/yahboomcar_ros2_ws/yahboomcar_ws/.venv`

```
yolo_env
```

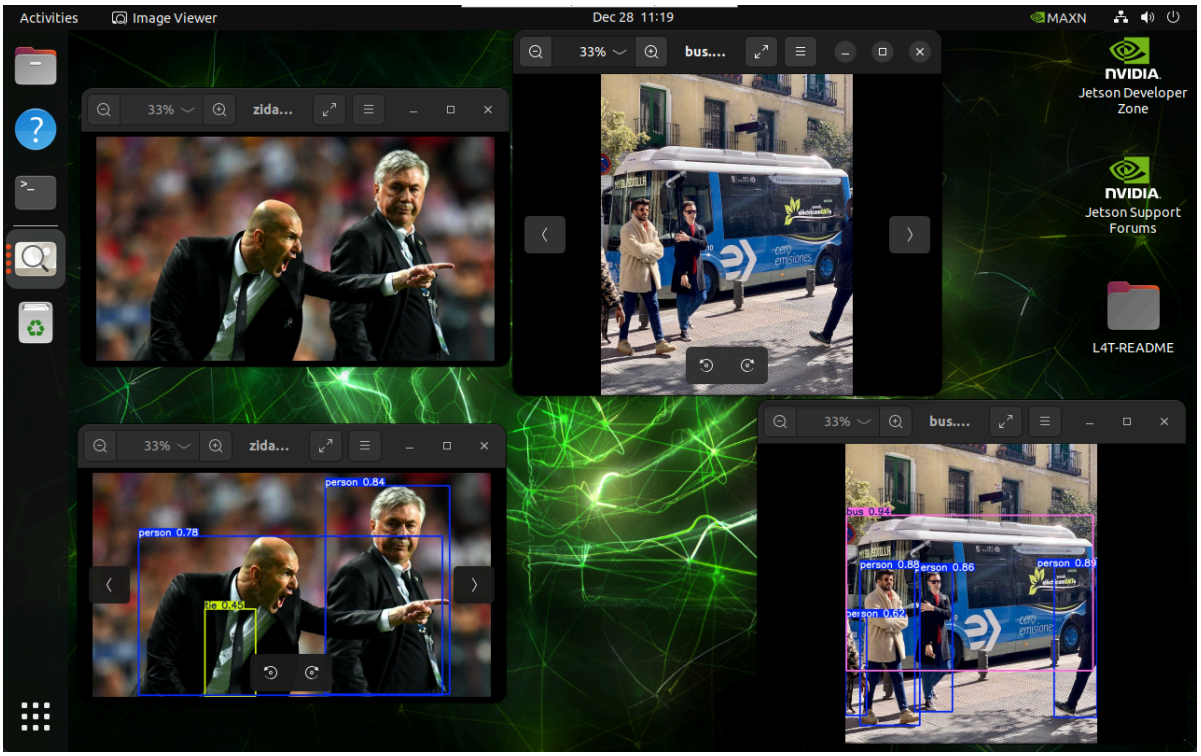
- Use yolo26n.pt to run prediction on a target image:

```
yolo predict model="$HOME/MODELS/yolo/yolo26n.pt"
source="$HOME/MODELS/yolo/assets/bus.jpg" save=True
```



Preview

With each run, the predict sequence number under this folder is incremented automatically:



2.3. Video Prediction

```
yolo_env
```

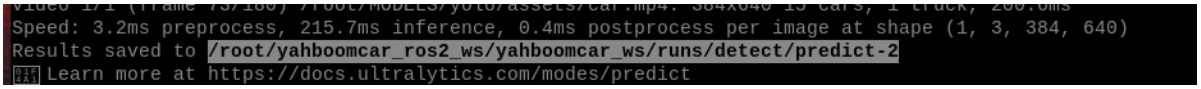
- Use yolo26n.pt to detect videos in the target folder and output the results:

```
yolo predict model="$HOME/MODELS/yolo/yolo26n.pt"
source="$HOME/MODELS/yolo/assets/car.mp4" save=True
```

```
video 5/5 (frame 389/414) /home/jetson/MODELS/yolov26/videos/street.mp4: 480x640 4 cars, 29.4ms
video 5/5 (frame 390/414) /home/jetson/MODELS/yolov26/videos/street.mp4: 480x640 3 cars, 29.4ms
video 5/5 (frame 391/414) /home/jetson/MODELS/yolov26/videos/street.mp4: 480x640 3 cars, 1 truck, 29.5ms
video 5/5 (frame 392/414) /home/jetson/MODELS/yolov26/videos/street.mp4: 480x640 3 cars, 1 truck, 29.4ms
video 5/5 (frame 393/414) /home/jetson/MODELS/yolov26/videos/street.mp4: 480x640 2 cars, 1 truck, 29.4ms
video 5/5 (frame 394/414) /home/jetson/MODELS/yolov26/videos/street.mp4: 480x640 2 cars, 29.4ms
video 5/5 (frame 395/414) /home/jetson/MODELS/yolov26/videos/street.mp4: 480x640 2 cars, 1 truck, 29.4ms
video 5/5 (frame 396/414) /home/jetson/MODELS/yolov26/videos/street.mp4: 480x640 2 cars, 1 truck, 29.4ms
video 5/5 (frame 397/414) /home/jetson/MODELS/yolov26/videos/street.mp4: 480x640 1 car, 2 trucks, 30.3ms
video 5/5 (frame 398/414) /home/jetson/MODELS/yolov26/videos/street.mp4: 480x640 4 cars, 3 trucks, 29.4ms
video 5/5 (frame 399/414) /home/jetson/MODELS/yolov26/videos/street.mp4: 480x640 2 cars, 3 trucks, 29.4ms
video 5/5 (frame 400/414) /home/jetson/MODELS/yolov26/videos/street.mp4: 480x640 4 cars, 2 trucks, 29.4ms
video 5/5 (frame 401/414) /home/jetson/MODELS/yolov26/videos/street.mp4: 480x640 2 cars, 3 trucks, 29.4ms
video 5/5 (frame 402/414) /home/jetson/MODELS/yolov26/videos/street.mp4: 480x640 4 cars, 3 trucks, 29.3ms
video 5/5 (frame 403/414) /home/jetson/MODELS/yolov26/videos/street.mp4: 480x640 3 cars, 3 trucks, 29.5ms
video 5/5 (frame 404/414) /home/jetson/MODELS/yolov26/videos/street.mp4: 480x640 6 cars, 2 trucks, 29.4ms
video 5/5 (frame 405/414) /home/jetson/MODELS/yolov26/videos/street.mp4: 480x640 6 cars, 1 truck, 29.5ms
video 5/5 (frame 406/414) /home/jetson/MODELS/yolov26/videos/street.mp4: 480x640 7 cars, 1 truck, 29.3ms
video 5/5 (frame 407/414) /home/jetson/MODELS/yolov26/videos/street.mp4: 480x640 2 cars, 1 truck, 29.4ms
video 5/5 (frame 408/414) /home/jetson/MODELS/yolov26/videos/street.mp4: 480x640 1 car, 1 truck, 29.4ms
video 5/5 (frame 409/414) /home/jetson/MODELS/yolov26/videos/street.mp4: 480x640 3 cars, 1 truck, 29.3ms
video 5/5 (frame 410/414) /home/jetson/MODELS/yolov26/videos/street.mp4: 480x640 4 cars, 3 trucks, 29.4ms
video 5/5 (frame 411/414) /home/jetson/MODELS/yolov26/videos/street.mp4: 480x640 4 cars, 2 trucks, 29.4ms
video 5/5 (frame 412/414) /home/jetson/MODELS/yolov26/videos/street.mp4: 480x640 6 cars, 3 trucks, 29.4ms
video 5/5 (frame 413/414) /home/jetson/MODELS/yolov26/videos/street.mp4: 480x640 3 cars, 3 trucks, 29.4ms
video 5/5 (frame 414/414) /home/jetson/MODELS/yolov26/videos/street.mp4: 480x640 3 cars, 3 trucks, 29.4ms
Speed: 3.9ms preprocess, 28.6ms inference, 1.1ms postprocess per image at shape (1, 3, 480, 640)
Results saved to /home/jetson/MODELS/output/runs/detect/predict3
💡 Learn more at https://docs.ultralytics.com/modes/predict
VS Code: view Ultralytics VS Code Extension ⚡ at https://docs.ultralytics.com/integrations/vscode
jetson@yahboom:~/MODELS/yolov26$
```

Preview

- Find the video file in the corresponding output path.



- Play the video (you can also directly open the video in the corresponding directory to view it):

```
ffplay <video_name>.avi
```

